PROITBRIDGE
Strive For Better Future

Cloud Deployment

(Project-Based Learning · For Professionals)

06 CHAPTERS | 12 PROJECTS | 18 TOOLS

12 HOURS

Project-Based Learning (8 Industries We Tap)

1. **SaaS & Startups** — Ship & scale web apps fast
2. **E-Commerce & Retail** — High-traffic store deployments
3. **Banking & FinTech** — Secure, compliant infrastructure
4. **Media & Streaming** — Scalable content delivery
5. **Healthcare** — Compliant cloud workloads
6. **Gaming** — Auto-scaling game backends
7. **IT Services** — Multi-cloud client delivery
8. **Enterprise DevOps** — CI/CD release automation

CHAPTER 1

Cloud Foundations & Deployment Mindset

From local apps to the cloud — how deployment really works

TOOLS

AWS · Azure · GCP · Cloud CLI · IAM

TOPICS

- What Cloud Deployment Is
- Regions & Availability Zones
- Shared Responsibility Model
- Cloud Pricing Basics
- IaaS · PaaS · SaaS
- Compute · Storage · Network
- Identity & Access (IAM)
- Free Tier Setup

PROJECTS

1. **Spin Up Your First Cloud VM**
2. **Host a Live Static Website**

→ Understand the cloud landscape → Deploy your first live workload

CHAPTER 2

Deploying on AWS

Ship and run real applications on Amazon Web Services

TOOLS

EC2 · S3 · RDS · Elastic Beanstalk · CloudWatch

TOPICS

- EC2 Instances
- S3 Object Storage
- Elastic Beanstalk
- Auto Scaling
- Security Groups & Keys
- RDS Managed Databases
- Load Balancing
- CloudWatch Monitoring

PROJECTS

3. **Deploy a Web App on EC2**
4. **Beanstalk App with RDS Database**

→ Run production apps on AWS → Scale and monitor workloads

CHAPTER 3

Deploying on Azure & GCP

Multi-cloud skills — App Service, VMs and managed runtimes

TOOLS

Azure App Service · Azure VMs · GCP Compute · Cloud Run

TOPICS

- Azure App Service
- Resource Groups
- Google Cloud Run
- Managed vs Self-Hosted
- Azure VMs & Storage
- GCP Compute Engine
- Cloud Storage Buckets
- Multi-Cloud Tradeoffs

PROJECTS

5. **Deploy an App on Azure App Service**
6. **Serverless Deploy on Cloud Run**

→ Deploy across Azure & GCP → Compare providers hands-on

CHAPTER 4

Containers with Docker

Package any app to run the same way everywhere

TOOLS

Docker · Dockerfile · Docker Hub · Docker Compose

TOPICS

- Why Containers
- Writing a Dockerfile
- Docker Hub Registry
- Docker Compose
- Images vs Containers
- Building & Tagging Images
- Volumes & Networking
- Multi-Container Apps

PROJECTS

7. **Containerize a Web Application**
8. **Multi-Service App with Compose**

→ Package apps into portable images → Run consistent environments anywhere

CHAPTER 5

CI/CD with Jenkins

Automate build, test and deploy from commit to production

TOOLS

Jenkins · Git · GitHub · Maven · Pipelines

TOPICS

- What CI/CD Solves
- Jobs & Builds
- Declarative Pipelines
- Credentials & Secrets
- Jenkins Setup
- Git Integration
- Build & Test Stages
- Plugins & Webhooks

PROJECTS

9. **Build Your First Jenkins Pipeline**
10. **Automated Test-and-Build Flow**

→ Automate the build-test cycle → Trigger deploys on every commit

CHAPTER 6

End-to-End Deployment Pipeline

Tie it together — a Dockerized app shipped to cloud via Jenkins

TOOLS

Jenkins · Docker · AWS · Azure · Monitoring

TOPICS

- Full Pipeline Design
- Pushing to a Registry
- Rollbacks & Versioning
- Secrets Management
- Containerized Builds
- Deploy to Cloud
- Monitoring & Logging
- Deployment Best Practices

PROJECTS

11. **CI/CD Pipeline that Deploys to AWS**
12. **Dockerized App Auto-Deploy**

→ Ship a full automated pipeline → Production-ready deployment skills

Strong Core NLP & Agentic AI knowledge is a must before this final Deployment phase.

CORE NLP → LLMs → RAG → AGENTIC AI → CLOUD DEPLOYMENT